

PABLO ALEJANDRO LÓPEZ SÁNCHEZ

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Data & AI Automation Engineer

PROFILE

Data & AI Automation Engineer specializing in the design and implementation of data architectures featuring centralized orchestration, MLOps infrastructure, and automation pipelines, reducing operational bottlenecks and optimizing strategic data-driven decision-making. Proven track record of building technical projects end-to-end, connecting abstract components into scalable, interconnected systems. Skilled at translating complex business requirements into scalable technical infrastructure.

SKILLS

Data Engineering:	Python (Polars, SQLAlchemy, FastAPI), SQL (SQL Server, PostgreSQL), ETL Pipelines, Web Scraping (Selenium, Scrapy).
AI & Agentic Systems:	Multi-agent development, spec-driven development, harness engineering, MCP (Model Context Protocol), Claude Code / Agent SDK, prompt engineering.
Infrastructure & Orchestration:	Dagster, Docker, SSH/SFTP, Linux Server, Git/GitHub, Data Flow Modeling.
Analytics:	Power BI (DAX, Power Query), KPI Analysis.
Business Strategy:	Project Management, Demand Planning, Inventory Optimization.
Languages:	English (B2-C1), Spanish (Native).

EXPERIENCE

Data & AI Automation Engineer Mercaldas – Manizales, Colombia	July 2025 – Present
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- Spearheading technical modernization efforts by migrating siloed, manual processes toward a centralized, automated, and scalable data architecture.*
- MLOps Infrastructure & Data Autonomy:** Pioneered the design and creation of an end-to-end MLOps infrastructure for autonomous demand forecasting. Achieved a **30%** improvement in forecast **accuracy** and a **3%** reduction in **stockouts** for high-priority products (A class), while eliminating reliance on third-party ERP providers.
 - Supply Chain Intelligence (AgTech/Livestock):** Developed an automated **web scraping** system to monitor livestock prices (beef, pork, poultry) and meat cuts, enabling data-driven negotiations for the company's meat production center.
 - Centralized Orchestration (Dagster):** Deployed and configured an orchestration platform on a dedicated **Linux** server to unify dispersed processes. Designed to automate manual executions and optimize failure monitoring, resulting in an estimated reduction of ~4 operational hours per week per user for the IT and Demand Planning teams.
 - Governance & DevSecOps:** Established the corporate development ecosystem on **GitHub**, implementing coding rules, version control, and **agentic workflows** using Claude AI to standardize internal software production.
 - Infrastructure Security:** Designed a technical roadmap to migrate legacy data flows (FTP) to secure protocols (**SFTP/SSH**). Planned the transition from daily full-refresh loads to hourly incremental extraction pipelines to enhance data availability, cybersecurity compliance and information integrity.

Business Analyst Latin America Sourcing – Remote	May 2025 – July 2025
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- Managed and analyzed commercial data using **Airtable** and **Stripe** to optimize negotiation processes with international suppliers.
- Analyzed client requests, documented commercial interactions, and generated operational insights. Automated supplier classification processes using custom Python scripts.

Business Intelligence Internship

Feb 2023 – Jul 2023

Industria Licorera de Caldas – Manizales, Colombia

- Developed Power BI dashboards for Operations and Foreign Trade departments, integrating data from Dynamics 365 to drive logistics efficiency.

EDUCATION

Bachelor of Business Administration (BBA)

Universidad Nacional de Colombia – October 2023

CERTIFICATES & COURSES

Full certificates list [here](#).

Google Data Analytics Professional Certificate

Google | Coursera

Databases for Data Scientists Specialization

University of Colorado Boulder | Coursera

Claude Code in Action

Anthropic Education

Data Analysis Tools

Wesleyan University | Coursera

REFERENCES

Luisa Soraya Santana Murcia

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